TBdecompile Reference Manual

Appendix B — Architecture and Gaps

What we know, what we know exists, and where the ? boxes are

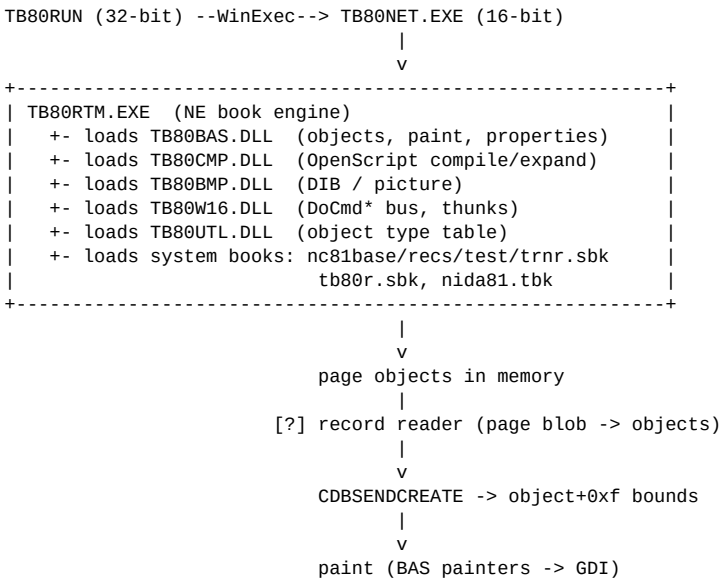
15 October 2026

How to read this

Green [+] marks a recovered fact. Red [?] marks a confirmed gap: we know the thing exists, we do not know the function or the offset.

1. Overall architecture

ToolBook II 8.1 is a 16-bit NE book engine (TB80RTM.EXE) with a 32-bit launcher (TB80RUN.EXE) and a satellite DLL set.



[+] TB80RTM is the engine. TB80BAS owns object create and paint. TB80UTL owns the type table.

[?] The function that walks a page blob and calls CDBSENDCREATE has not been named.

2. Book open path

```
Tbload / TB80RUN
     |
     v
find ZZZTBKBOOKLOC (file offset of book data)
     |
     v
JBO signature check (virtual method at *param_1+8)
     |
     v
[?] page structure allocation (virtual method)
     |
     v
CDBFILEOPEN (BAS ordinal 8)
     |
     v
page objects appear in memory (80 65 records)
     |
     v
[?] for each page: walk nested 14 00 00 chunks
     |
     v
CDBSENDCREATE (BAS ordinal 51)
     |
     v
object exists, bounds at +0xf
```

[+] ZZZTBKBOOKLOC is searched by Tbload. JBO is checked. CDBFILEOPEN and CDBSENDCREATE are named.

[?] The virtual method that allocates page structures, and the walker that feeds CDBSENDCREATE, are unnamed.

3. Object create path (recovered)

```
CDBSENDCREATE (12c8:00bc, Ordinal_51)
     |
     +- param_5 == 0x1014 / 0x1015 --> FUN_12c8_0850
     |                                      |
     |                                      v
     |                                 FUN_12c8_0e32 (allocator)
     |                                      |
     |                                      v
     |                                 FUN_12c8_0ef8 (field copier)
     |                                      |
     |                                      v
     |                                 object+0xf = bounds
     |
     +- param_5 == 0x1021 / 0x1022 --> FUN_12c8_0a86
     +- param_5 == 0x3018-0x301b --> CDBCHECKBOUNDDTL + FUN_12c8_0e08
     +- else --> Ordinal_115
```

[+] Object is 0x76 bytes. Bounds are four words at +0xf/+0x11/+0x13/+0x15.

[+] FUN_12c8_0ef8 copies source+0x3e/+0x40/+0x42/+0x44 to object+0x17/+0x19/+0x1b/+0x1d.

[?] The source structure that FUN_12c8_0ef8 reads is not the on-disk 14 00 00 record. An expansion step sits between them.

4. Object layout (partial)

offset	size	meaning	status
-2	word	type code (0x11 = arc)	[+]
+0x0f	word	left (twips)	[+]
+0x11	word	top	[+]
+0x13	word	right	[+]
+0x15	word	bottom	[+]
+0x17..	words	copied from source+0x3e..	[+]
+0x1f	byte	flags / BitBlt select	[+]
+0x30	word	polygon point count	[+]
+0x32	ptr	polygon point array (twips)	[+]
+0x3b	word	GDI handle	[+]
+0x25..	pairs	arc geometry (four twip pairs)	[+]
+0x??	??	picture resource id	[?]
+0x??	??	script / handler pointer	[?]
+0x??	??	parent / group link	[?]

[?] Picture id, script pointer, and parent link have not been located in the object.

5. Paint path

```
WM_PAINT (RTM)
     |
     v
CLIENTTOPAGE
     |
     v
CDBPAINTCLIENT (BAS Ordinal_26)
     |
     v
type dispatcher (BAS segment ~0x7E)
     |
     +- polygon --> PaintPolygon (seg 0x82, 1408:0376)
     |                 reads object+0x30 count, +0x32 points
     |                 ScaleTwipsToScreen (seg 2, 1008:01dd)
     |                 GDI!POLYGON
     |
     +- rectangle --> FUN_1410_0066 (seg 0x83)
     |                 reads bounds at +0x0f
     |                 GDI!RECTANGLE
     |
     +- arc --> PaintArc (seg 0x7B)
     |            four twip pairs at +0x25
     |            GDI!ARC
     |
     +- curve --> [?] stroke walker
     |              points exist; control structure unknown
     |
     +- picture --> [?] property -> TB80BMP ordinal 3 -> DIB
                      object+0x3b observed as GDI handle
```

[+] ScaleTwipsToScreen: screen = (stored * 96 + 720) / 1440. On the observed DS:01A6=96, this is stored/15.

[?] Curve stroke control structure (P420) and picture-id property are unnamed.

6. On-disk records

```
Page object:  80 65 00 05 00 02 02   (marker)
              name is 32-byte field
              property block at +43 starts 14 00 00 01

Widget:       14 00 00 <subtype>     nested inside page
              subtype 0 + +0x0c=04 00  -> polygon / content rect
                fill color dword at +4
                point count word at +0x1a
                x,y twip pairs at +0x22
                screen = twips / 15

              bounds-only rect:
                four twip words + fill
                marker pattern 01 00 00 04 nearby

OpenScript:   Text\0\0 property
              8 bytes, then u16 length, then string
              handlers are bytecode; keyword table is resource 0x77
```

[+] Polygons and content rectangles have been parsed directly from the page blob.

[?] The runtime reader that turns these chunks into objects is not the same code that our tools use. It has not been located.

7. OpenScript path

```
on disk: handler bytecode in page blob
     |
     v
resource 0x77 (keyword offset table)
     |
     v
COMPILEREXPANDTEXT (TB80CMP)
     |
     v
source text  ("go to page \"P400\"", "send next", ...)
     |
     v
[?] expression evaluator (FUN_1008_2d31 is the FPU path)
     |
     v
runtime dispatch (goto, send, assign, if/else, step)
```

[+] Decoder recovers readable handlers. Common subset runs in the clean-room engine.

[?] Full expression evaluation and the compile-time token map are only partially recovered.

8. System books

```
nc81base.sbk   object type table, menus
nc81recs.sbk   session / response recording
nc81test.sbk   quiz machinery
nc81trnr.sbk   trainer / hardware probe
tb80r.sbk      Asymetrix reader log / help
nida81.tbk     shared shell (splash, menus, copyright)

nc81int.dll    47 trainer calls (130E, TD2000, 5050, 205E)
nc81rec.dll    RecStartSession, RecWriteResponse, ...
```

[+] System books are resource containers. Logic is in the DLLs, not in OpenScript.

[?] The exact handoff from a lesson script to nc81int / nc81rec is only partially traced.

9. Confirmed gaps (the ? boxes)

These are the places where further debugger work or forced disassembly is expected to pay off.

Gap	What exists	Where to look
Page blob walker	Walks 14 00 00 chunks, calls CDBSENDCREATE	Break on CDBSENDCREATE, walk stack
Source expansion	Turns disk record into the structure FUN_12c8_0f8a reads	Compare buffer on stack with page blob
Curve strokes	P420 point runs; control structure unknown	Trace BAS segment 0x5B or type dispatcher
Picture id	Property that TB80BMP ordinal 3 resolves	Break on ordinal 3, inspect object
Script pointer	Where the handler lives in the object	Break on script dispatch
Parent / group	How relative bounds become absolute	Trace CLIENTTOPAGE or bounds setter
Expression eval	Full OpenScript math beyond the FPU path	Compile+expand a short script

10. Pseudocode for the missing reader

Working model. Not recovered. Fits the evidence.

```
function LoadPage(page_blob):
    offset = 0
    while offset < length(page_blob):
        chunk = page_blob[offset]
        if chunk.marker != 0x14:          # length-prefixed, not searched
            break
        subtype = chunk[3]
        if subtype == 0 and chunk[+0x0c] == 0x0400:
            # polygon or content rect
            bounds = extract_bounds(chunk)   # [?] expansion
            obj = CDBSENDCREATE(type=0x1014, bounds)
            if has_points(chunk):
                SetPointArray(obj, chunk[+0x1a], chunk[+0x22])
            set_fill(obj, chunk[+4])
        elif is_picture(chunk):
            # [?] picture id property
            obj = CDBSENDCREATE(...)
            set_property(obj, PICTURE_ID, chunk.resource_id)
        elif is_script(chunk):
            # handler bytecode
            attach_handler(obj, chunk.bytecode)
        offset += chunk.length
    return objects
```

The expansion step (disk record to source structure) is the single largest unknown. Everything else in the create path is named.

11. What this means for a clean-room reader

A modern reader does not need the unnamed functions. It needs the disk layout, which is partially recovered.

[+] Page objects are findable by the 80 65 marker.

[+] Polygons and content rectangles can be drawn from the 14 00 00 chunks.

[+] OpenScript handlers can be expanded to source.

[+] System books supply type names and menus.

[?] Absolute positioning of nested objects, picture binding, and curve stroke order still require either the missing reader or a working theory that matches the TBX expansion.

The ? boxes are the research targets. The [+] boxes are the scaffolding a clean-room implementation can already stand on.